**Scope of Work: Develop an Arduino based control board and sketches to run a RC cart wirelessly with following parameters:**

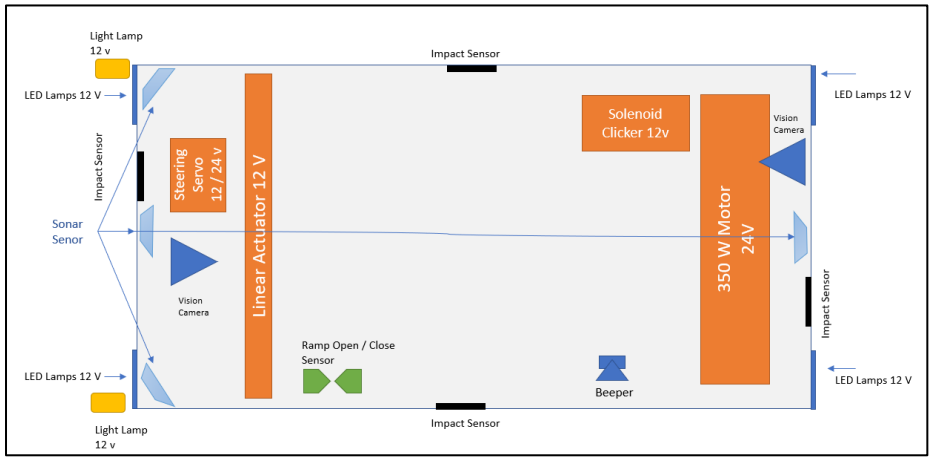
1. **RC Communication 2.4 GHZ Remote**
2. **Camera System (FPV Camera)**
3. **Obstacle Detection (no avoidance functionality) only collision prevention**

Sketches will be for main Arduino board to control motors and a remote Arduino based controller system.

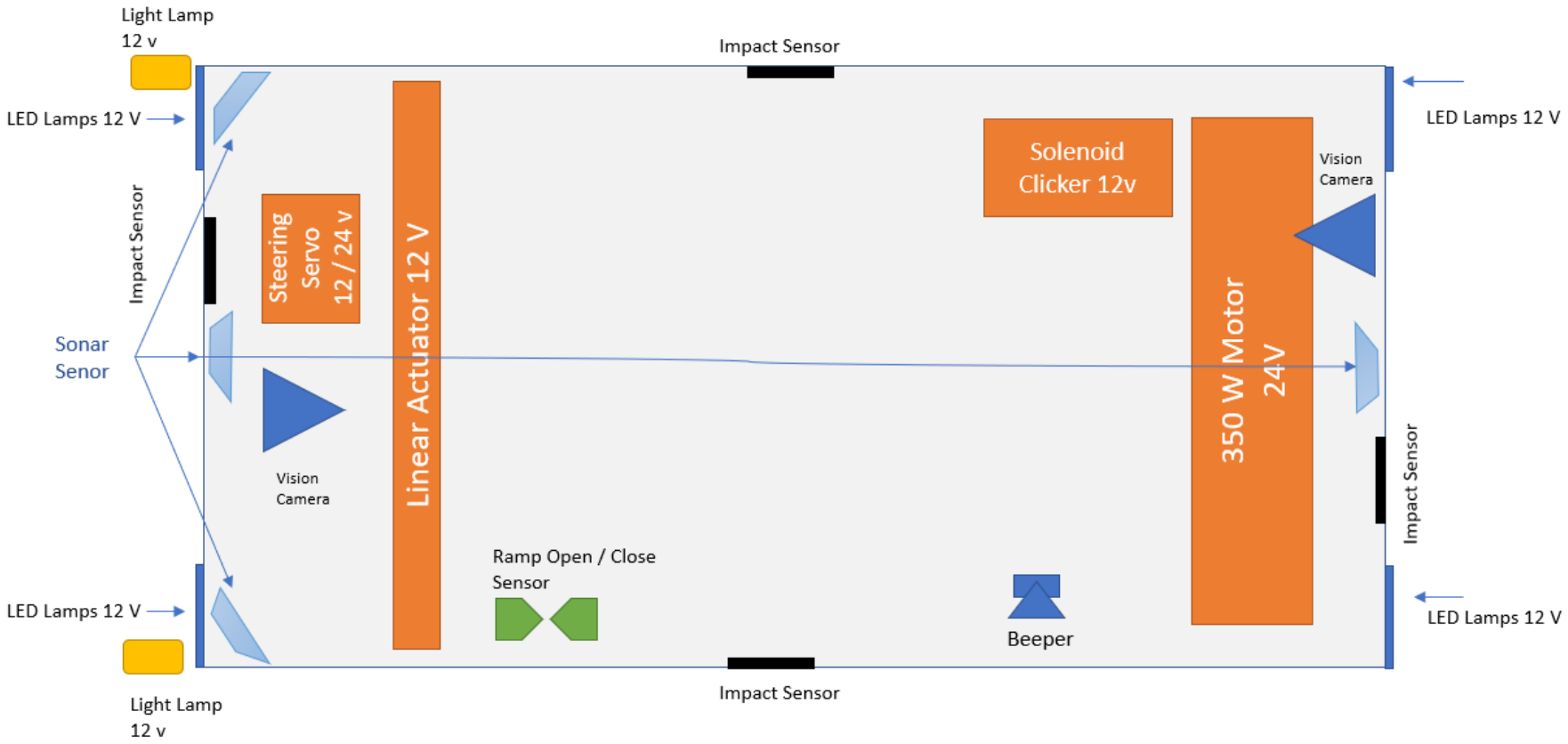
**Carrier: The cart system that has the following components:**

- LED Lamps X 4
- Sonar Sensors x 4
- Impact Sensor x 4
- Vision Camera x 2
- Steering Servo x 1
- Linear Actuator x 1
- Solenoid x 1
- Motor Transaxle x 1
- Beeper x 1
- Light Lamp x 2

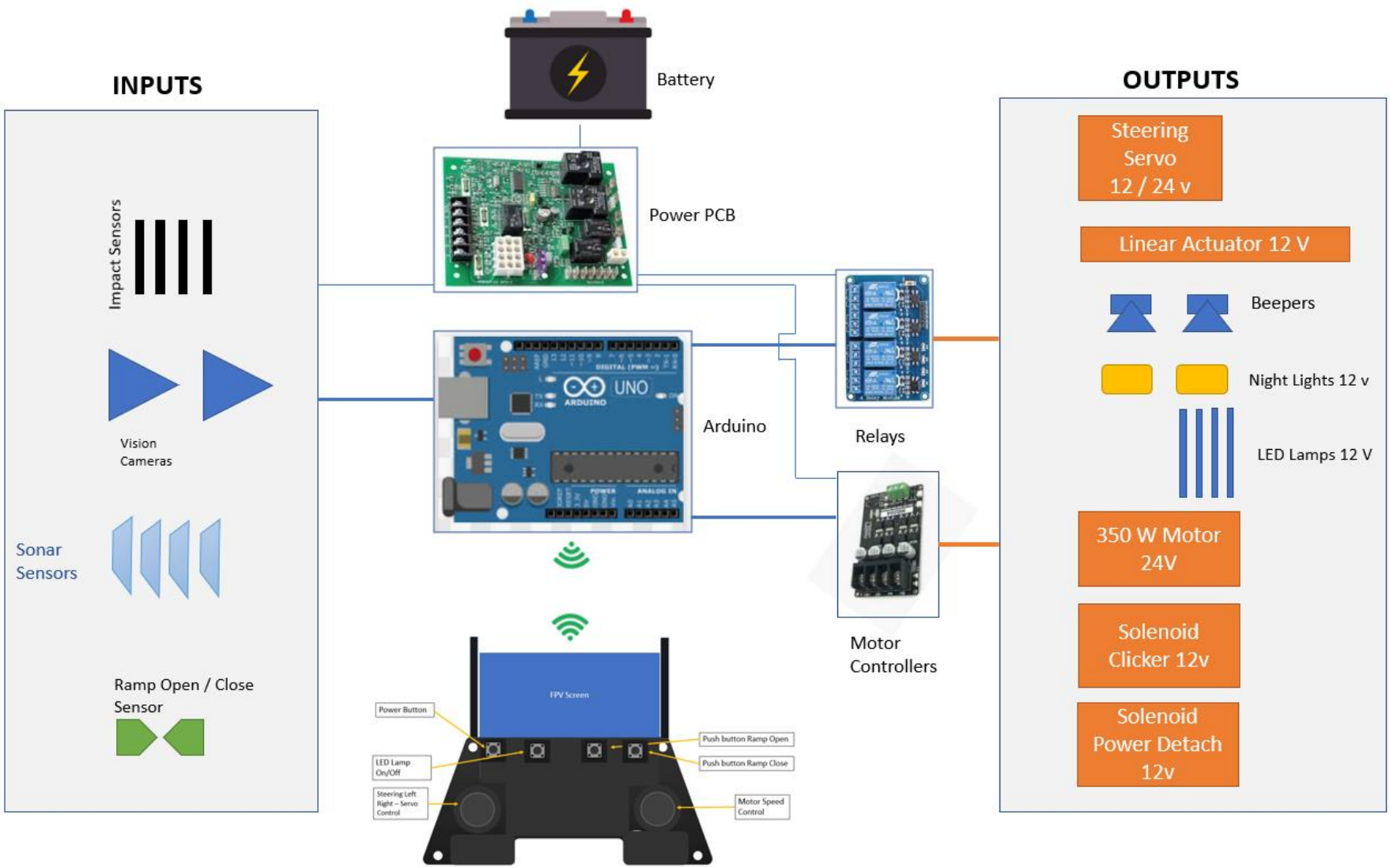
Layout



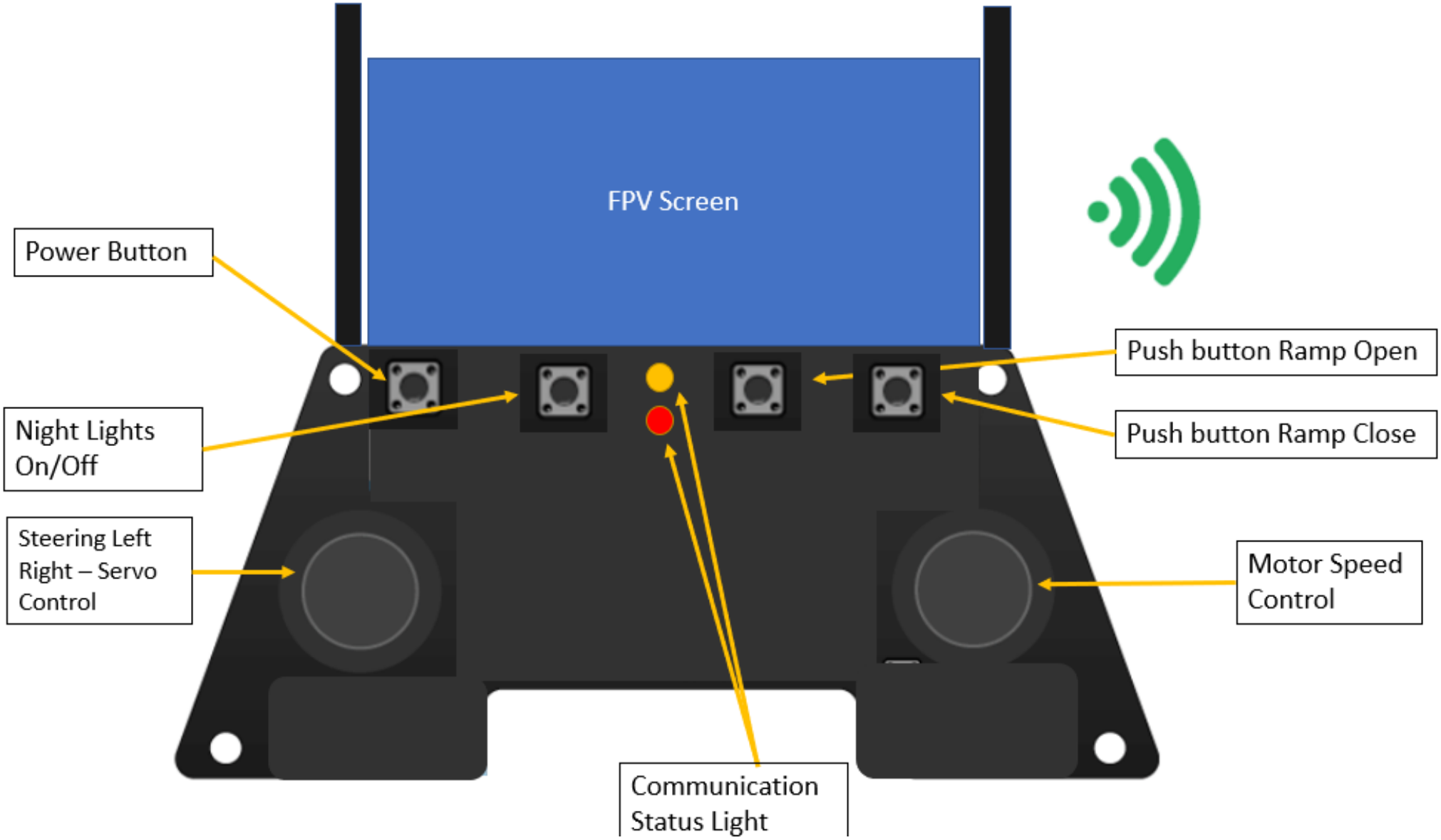
## General Architecture and layout of sensors, motors and devices



# I/O Map


INPUTS
Impact Sensors
Vision Cameras
Sonar Sensors
Ramp Open / Close Sensor
Battery
Power PCB
Arduino
Relays
Motor Controllers
FPV Screen
Power Button
LED Lamp On/Off
Steering Left Right – Servo Control
Push button Ramp Open
Push button Ramp Close
Motor Speed Control
OUTPUTS
Steering Servo 12 / 24 v
Linear Actuator 12 V
Beepers
Night Lights 12 v
LED Lamps 12 V
350 W Motor 24V
Solenoid Clicker 12v
Solenoid Power Detach 12v

## Remote Layout



**Function ` the Cart:** The cart should work wirelessly with remote control.

1. Power Button: This will power on the PCB and will provide aux power to all components.
2. Once the communication between the remote and cart is established only then the user should be able to control the cart.
3. The sensors (impact sensor and sonar) should be able to detect obstacle and should cut the power to drive motor if:
    a. Obstacle detected by sonar – disallow movement in obstacles path
    b. Impact Sensor activated – disallow movement for 10 seconds, repeated impact sensor hit more than 3 times should disallow motor movement for two minutes and display message on screen – Obstacle detected.
    c. Night Lights: Turn on and off the lamps
4. Linear Actuator: controls the ramp function any time stall current exceeds threshold then it should display notification that ramp is stuck
5. I know of FPV camera that transmits video feed over 2.4 GHz channel, somehow notifications will need to be displayed on the screen – all notifications are outlined on logic chart.
6. If the cart is outside the receiver range, it should stop and wait for signal
7. Movement of cart should turn on LED lights and Beeper.

All Parts are outlined in the Appendix Section and specs will be provided.

**Tasks:**

**Task 1: Develop library (if needed) and code (sketches) for Arduino to control the following:**

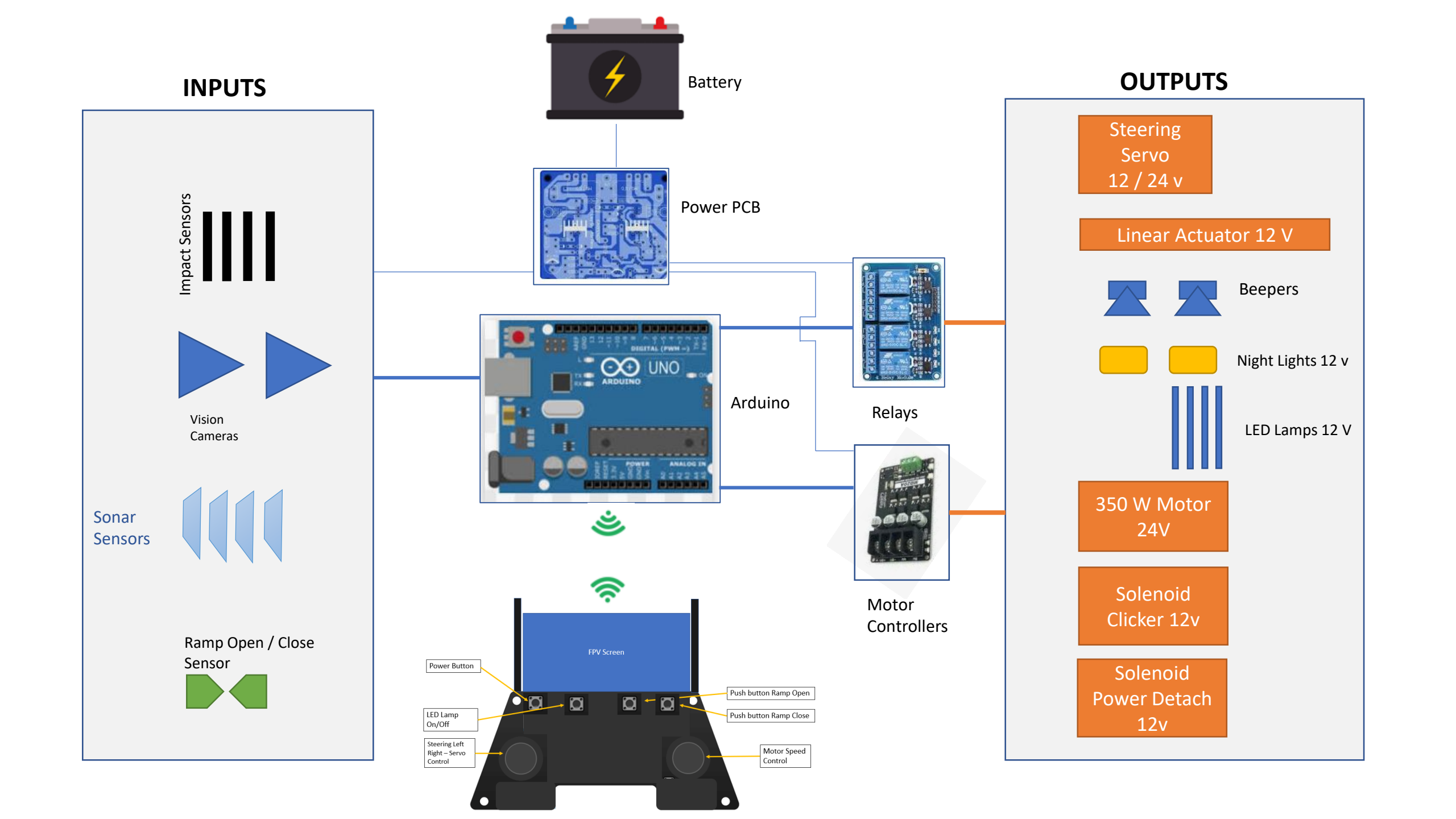
1. Power on the System
2. Drive 350 W brushed motor with motor controller
3. Steer the cart using servo motor with motor controller
4. Provide power feed to vision cameras
5. Engage Sonar sensor to detect and perform collision avoidance
6. Open and close linear actuator based on remote command
7. Click electric soleniod using remote command.
8. Self detach the charging cable

**Task2 :Design and code RF Remote Control system**

**Task 3: Provide instructions and guidance for uploading sketches, debugging any information and offer troubleshooting to make the arduino uno and remote work. Provide assitance with correct wiring and working of the units.**

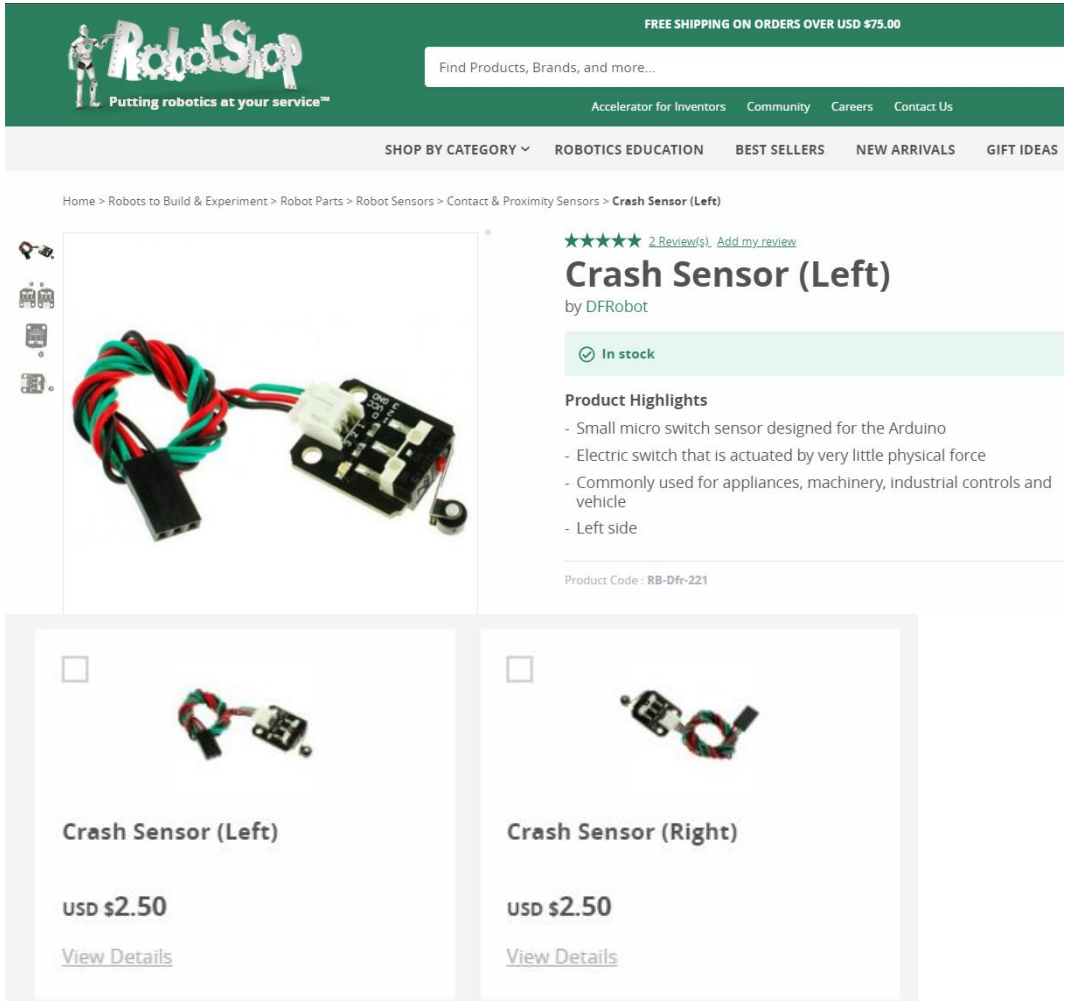
**Deliverables:**

1. Wiring sketch for main Arduino Board
2. PCB board that plugs in on top of arduinio with required
3. Sketch for Remote controller
4. PCB board and design for remote controller
5. Wiring diagram and sketch upload files and sequence for programming
6. Parts listing needed to assemble everything
7. Provide assistance and trouble shooting as needed to make the unit work
8. Offer Enhancements and Efficiency to design


INPUTS
Battery
Power PCB
Impact Sensors
Vision Cameras
Sonar Sensors
Ramp Open / Close Sensor
Arduino
Relays
Motor Controllers
FPV Screen
Power Button
LED Lamp On/Off
Steering Left Right – Servo Control
Push button Ramp Open
Push button Ramp Close
Motor Speed Control
OUTPUTS
Steering Servo 12 / 24 v
Linear Actuator 12 V
Beepers
Night Lights 12 v
LED Lamps 12 V
350 W Motor 24V
Solenoid Clicker 12v
Solenoid Power Detach 12v

## INPUTS

Impact Sensors ||||




https://www.robotshop.com/en/crash-sensor-left.html

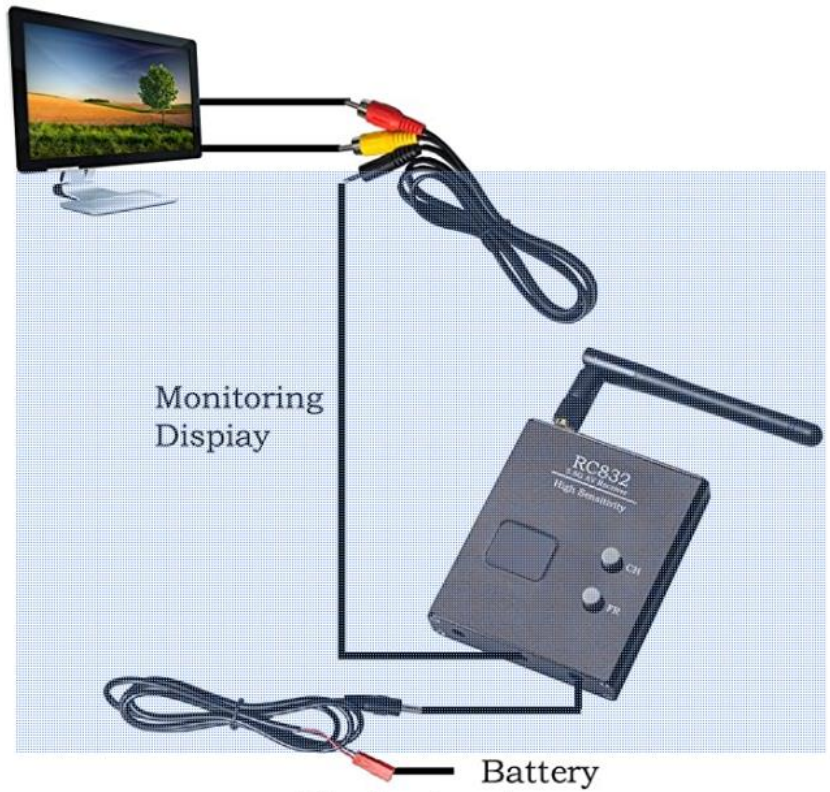
# INPUTS

Vision
Cameras

‹ Back to results



## Application example:




Roll over image to zoom in

**Features:**
40 channels: 5 bands and all frequencies compatible
Two switching buttons: easy to change frequency channels and bands
Two digital display: real-time positioning which frequency band and which
Power off memory: replay the very last frequency band and channel

**Specification for TS832 Transmitter:**
Modulated: Wideband FM Modulation
40 channels: Cover A, b, E, F, r bands
Supply current: 220 mA
Frequency: 5.8GHz 5 bands 40 channels, with Raceband: 5658-5917MHz
Transmitting distance: > 3000m (open area)
Transmitter connector:RP SMA female
Antenna connector: RP SMA male
Audio carrier Frequency: 6.5 MHz
Audio Input Level: 0.5 ,2.0 Vp-p
Audio Input Impedance: 10K Ohm
Output Impedance: 50 Ohm
Operating Voltage:7-16V
Operating Temperature: -10 +85 ℃
Output Power: 27-28 dBm
Video Format: NTSC/PAL
Video Band Width: 0-8.0 MHz
Video Input Level: 0.8,1.0,1.2 Vp-p
Video Input Impedance: 75 Ohm
Weight: 22g
Dimension (L×W×H): 2.12×1.23×0.39 inches

**Specification for RC832 Receiver:**
Power Supply: DC 12V
Working Frequency: 5.8GHz
Sensitivity: ≤-90 dBm
Type Standard: NTSC/PAL
Consumption Current: 200mA, Max
Receiver connector: RP SMA female
Antenna connector: RP SMA male
Available Channel: 40CH
Antenna Gain: 2dbi
Audio Carrier: 6.5MHz
Audio Output Level: 1.0 Vp-p Typ 10 K Ohm
Antenna impedance: 50 Ohm
Antenna Input Impedance: 50 Ohm TYP
Video impedance: 75 Ohm
Video Format : NTSC/PAL
Video Output Level: 1.0 Vp-p Typ 75 Ohm
Dimension: 80x65x15mm
Weight: 85g
Dimension (L×W×H): 3.15×2.56×0.59 inches

https://www.amazon.com/Wolfwhoop-5-8GHz-Wireless-Transmitter-Receiver/dp/B01LQ04E5E/ref=sr_1_5?dchild=1&keywords=fpv+camera+receiver&qid=1593574698&sr=8-5

## INPUTS

Sonar Sensors





## Specifications

- Operating voltage: DC3.0-5.5 V
- Static current: >8 mA
- Operating current: 30 mA
- Operating range: 20 cm ~ 6 m
- Resolution: 1 mm
- Detecting Angle: < 75°
- Operating Temperature: -20 ~ 70 °C
- Cable Length: 2.5 m
- Weight: 54 g

## Useful Links

**Zip file**

- Sample code

**Website**

- Github

★★★★★ 5 Review(s) Add my review

# Weatherproof Ultrasonic Sensor w/ Separate Probe

by DFRobot

In stock

**Product Highlights**

- Weatherproof Ultrasonic Sensor w/ Separate Probe
- Features detectable ranges from 0.2 to 6m
- Offers 1 mm resolution
- Works at an input voltage of 3.3 to 5V DC

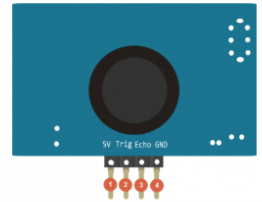
Product Code : **RB-Dfr-720**

## Description

- Weatherproof Ultrasonic Sensor w/ Separate Probe
- Features detectable ranges from 0.2 to 6m
- Offers 1 mm resolution
- Works at an input voltage of 3.3 to 5V DC

The **Weatherproof Ultrasonic Sensor w/ Separate Probe** is suitable for outdoor applications such as car reversing sensors, security alarms, industrial inspection, etc.

As most other ultrasonic distance sensors aren't waterproof it can be a problem as you need your project to withstand the elements outdoors. No need to worry anymore! This sensor is weatherproof and the latest Version 2.0 can work at 3.3-5 V, while it provides several modes to choose from.




| LABEL | NAME | FUNCTION |
|---|---|---|
| 1 | 5V | Power Input + |
| 2 | Trig | Trigger Signal |
| 3 | Echo | Echo Signal |
| 4 | GND | GND |

**Requirements**

- 1 x Arduino UNO
- 1 x Battery capacity indicator
- Arduino IDE V1.6.9 Software

https://www.robotshop.com/en/weatherproof-ultrasonic-sensor-separate-probe.html?utm_source=google&utm_medium=surfaces&utm_campaign=surfaces_across_google_usen&gclid=Cj0KCQjw6PD3BRDPARIsAN8pHuF99LbEp96eODb06f457DKcWD6koyLSoLVvCegX-RrvwFdjpnej_c0aAkeDEALw_wcB

## INPUTS

Ramp Open / Close Sensor



Home > Robots to Build & Experiment > Robot Parts > Robot Sensors > Contact & Proximity Sensors > **Micro Contact / Limit Switch**



Click to enlarge

★★★★★ 4 Review(s) Add my review

# Micro Contact / Limit Switch

by Tamiya

In stock

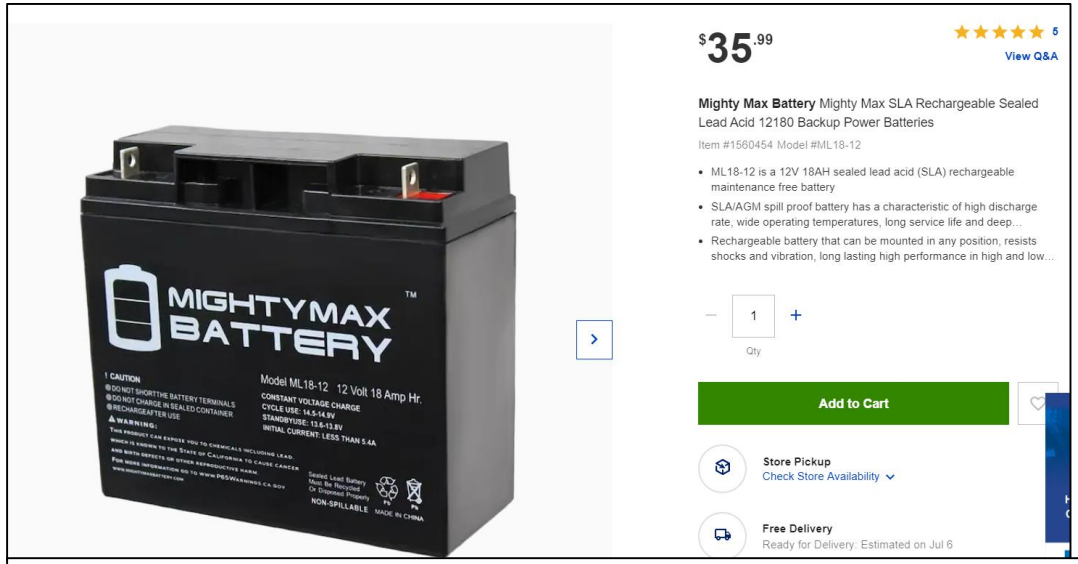
**Product Highlights**

- 5A Micro Switch
- Uses 6V and up to 4A capacity
- Forward/reverse Capacity
- Functions: On/off

Product Code : **RB-Tam-71**

https://www.robotshop.com/en/micro-contact-limit-switch.html






https://www.lowes.com/pd/Mighty-Max-Battery-Mighty-Max-SLA-Rechargeable-Sealed-Lead-Acid-12180-Backup-Power-Batteries/1001107752?cm_mmc=shp-_-c-_-prd-_-hdw-_-google-_-pla-_-214-_-sosbatteries-_-1001107752-_-0&placeholder=null&gclid=Cj0KCQjw6PD3BRDPARIsAN8pHuGpXcaMEVJ3IZ06Coi7lM9YHBwDpnudezZRSVzAa1SL3Jr2LNQARNsaAohSEALw_wcB&gclsrc=aw.ds

Delivering power when you need it, the mighty max ML18-12 12 Volt 18 AH uses a state of the art, heavy-duty, calcium-alloy grid that provides exceptional performance and service life in both float and cyclic applications. The ML18-12 is an absorbent glass mat (AGM) technology with a valve regulated design, that can be used in enclosed and indoor environments without leaking or maintenance, providing a superior performance for thousands of models. Mighty max SLA batteries are utilized in a wide variety of applications including; consumer electronics, electric vehicles, engine starters, golf carts, hunting, lawn and garden tools, medical mobility, motorcycles, power sports, portable tools, solar, toys and hobby, access control devices, emergency lighting, security and more.

- ML18-12 is a 12V 18AH sealed lead acid (SLA) rechargeable maintenance free battery
- SLA/AGM spill proof battery has a characteristic of high discharge rate, wide operating temperatures, long service life and deep discharge recover
- Rechargeable battery that can be mounted in any position, resists shocks and vibration, long lasting high performance in high and low temperatures
- Terminal is nut and bolt listing is for the battery and screws only no wire harness or mounting accessories included
- MIGHTY MAX SLA batteries are utilized in a wide variety of applications including,electronics, electric vehicles, golf carts, medical mobility, motorcycles, power sports,solar, toys and hobby, access control devices, emergency lighting, security and more

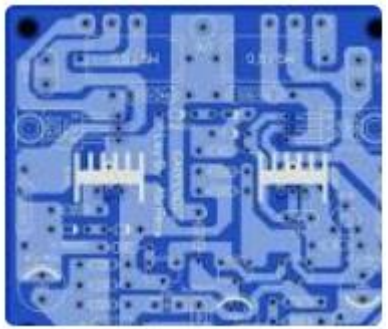
CA Prop 65
PDF

CA Residents: ⚠ Prop 65 Warning(s)

**Specifications** +

**Rating & Reviews** ★★★★★ 5 +

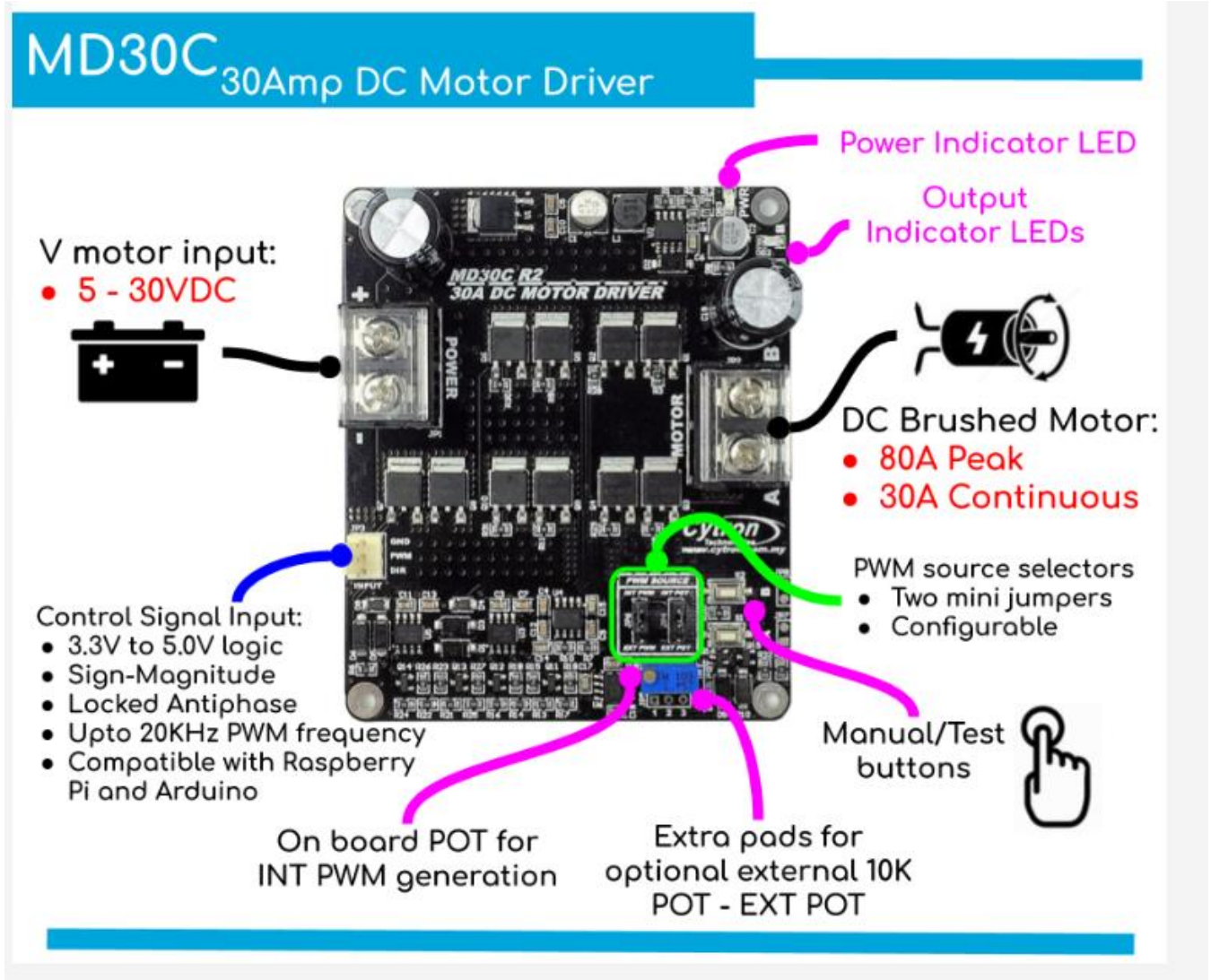
**Community Q & A** +



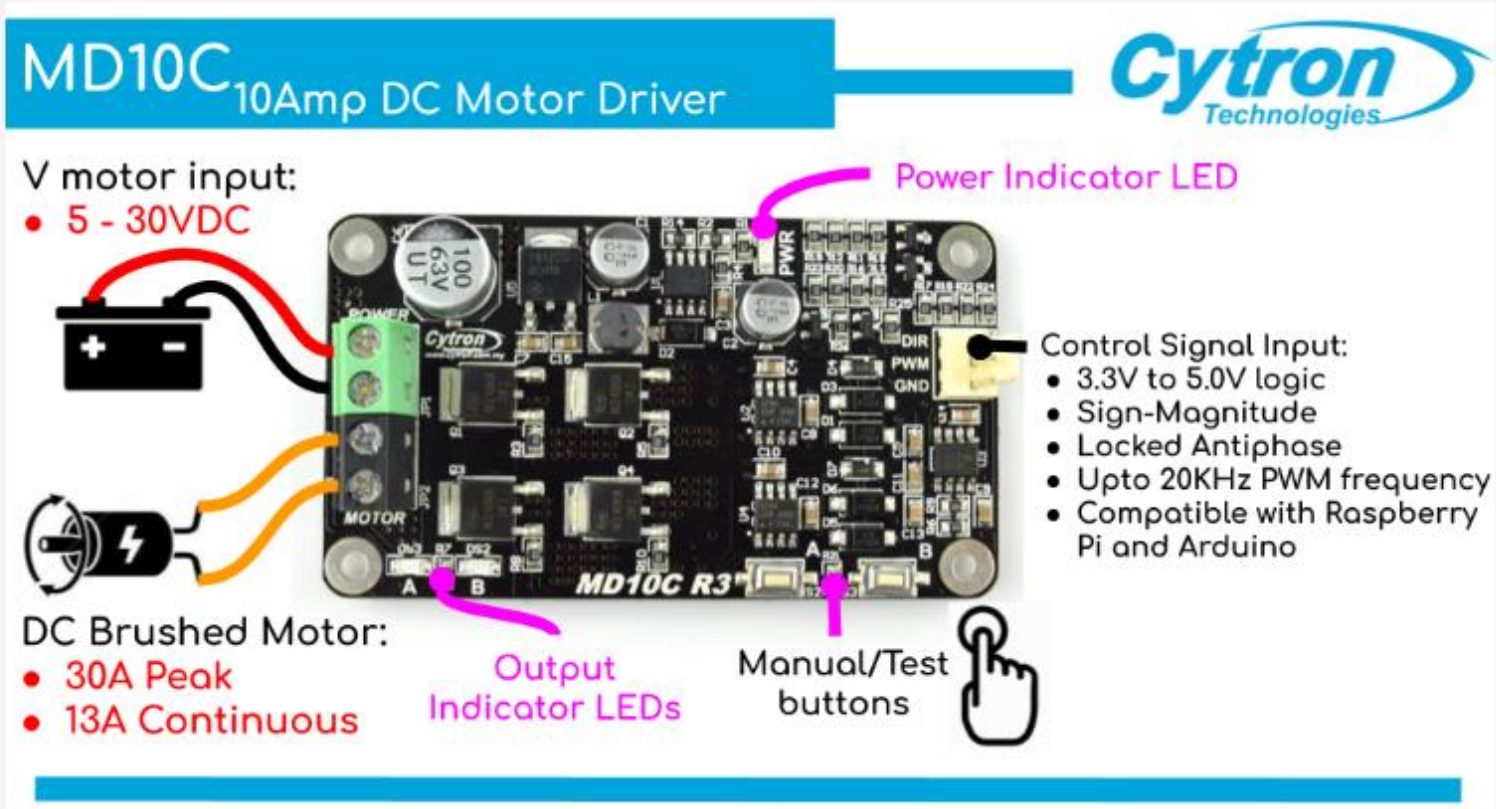
This will be designed by you to supply power to all the required components and units.




This is Arduino Uno – please let me know if we need a different unit for the task.




https://www.cytron.io/p-30amp-5v-30v-dc-motor-driver




Overview of MD10C

https://www.cytron.io/p-10amp-5v-30v-dc-motor-driver





https://www.aliexpress.com/item/32657994793.html

## OUTPUTS

Steering
Servo
12 / 24 v




https://www.amazon.com/Progressive-Automations-Feedback-Linear-Actuator/dp/B00Q74D9ZG/ref=sr_1_2_sspa?crid=34TCAJLQZ9E15&dchild=1&keywords=linear+actuator&qid=1593637945&sprefix=linear+actuat%2Caps%2C159&sr=8-2-spons&psc=1&spLa=ZW5jcnlwdGVkUXVhbGlmaWVyPUFOTDBCTUtYSzUzV08mZW5jcnlwdGVkSWQ9QTAzMjQ5NzQzSFdEUFM5QUoyNU1SJmVuY3J5cHRlZEFkSWQ9QTAzODAyMzIyMjNZSUlCS09YNVBPJndpZGdldE5hbWU9c3BfYXRmJmFjdGlvbj1jbGlja1JlZGlyZWN0JmRvTm90TG9nQ2xpY2s9dHJ1ZQ==

## OUTPUTS

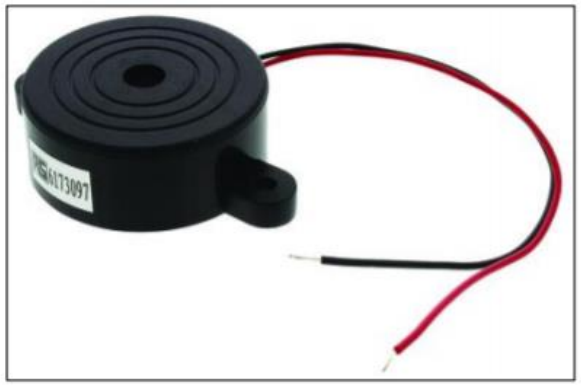
Linear Actuator 12 V

Professionally approved products.

Datasheet

# 12V dc 96dB Continuous Tone Buzzer

RS Stock number 617-3097



https://www.alliedelec.com/m/d/a08deaa282816228ec6d7f6e542b6e25.pdf

**OUTPUTS**



Beepers




Nilight Led Light Bar 2PCS 18w Spot Driving Fog Light Off Road Lights Boat Lights driving lights Led Work Light SUV Jeep Lamp,2 years Warranty

by Nilight

1,880 ratings | 196 answered questions

Price: $17.40 **FREE Shipping** on your first order. Details & FREE Returns

Get $50 off instantly: Pay $0.00 ~~$17.40~~ upon approval for the Amazon Rewards Visa Card. No annual fee.

Available at a lower price from other sellers that may not offer free Prime shipping.

Size: **2PCS 18W LED Lights**

| | | |
|---|---|---|
| **2PCS 18W LED Lights** $17.40 | 2PCS 42W LED Pod $28.99 | 2PCS 54W White/Amber Lights $18.62 |
| 4PCS 18W LED Lights $26.09 | | |

- SUPER BRIGHT LIGHT: High-quality LED chips emit very bright and white light, PMMA lens with high light transmittance, clear view.
- FURTHER IRRADIATION DISTANCE: Concentrated spot beam illuminates to further distance, lighting up the forward dark road.
- Enjoy the rounded shoulder design and durability of the glass, and reuse your bottles over and over. Great for essential oils, perfume and other oils and liquids, Dispense liquids precisely.
- SUPERIOR QUALITY WIRES: Good electrical conductivity, high temperature and melting resistant, outstanding bending performance.
- PACKAGE INCLUDE: 2PCS 18W Spot LED Work Lights

https://www.amazon.com/Nilight-Driving-Lights-driving-Warranty/dp/B07217HPV5/ref=sr_1_14?dchild=1&keywords=fog+lamps+small&qid=1593638478&sr=8-14

## OUTPUTS

Night Lights 12 v




https://www.amazon.com/Universal-Motorcycle-Indicator-Cruiser-mounting/dp/B011DXKB42

## OUTPUTS

LED Lamps 12 V

# CT5

Transaxle Drive Unit



## Specification

| | |
|---|---|
| Reduction Ratio | 17:1 / 19.7:1 |
| Vehicle Capability | 100kg (220lb) |
| Load Capability | 110kg (250lb) |
| Motor Maximum Rated Power | 350W |
| EMB Torque | 4 N.m |

## Applications

Mobility Scooter | Electric Cart | Cleaning Machine | Service Robot | Automated Guided Vehicle (AGV)

https://ctengineering.com.tw/download/CT5.pdf

**OUTPUTS**

350 W Motor
24V

## 412

adafruit



Enlarge

Images are for reference only
See Product Specifications

| | |
|---|---|
| **Mouser #:** | 485-412 |
| **Mfr. #:** | 412 |
| **Mfr.:** | Adafruit |
| **Customer #:** | Customer # |
| **Description:** | Adafruit Accessories Small Push-Pull Solenoid |
| **Datasheet:** | 412 Datasheet (PDF) |

https://www.mouser.com/datasheet/2/737/Adafruit_05132020_412-1858431.pdf

**OUTPUTS**

Solenoid Clicker 12v

Solenoid Power Detach 12v